The Ethics of Automated Research

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Automation journeys

Which automation journey is your laboratory on? Which journey best fits your current workflow? Determine the right solutions to automate your laboratory.

Instrument loading journey

Need to automate instrument sample loading and unloading steps?

Key technologies & workflows:

→ Plate reading
→ Sample prep

Simple workflow journey

Need to automate a set of benchtop instruments or a simple process?

Key technologies & workflows:

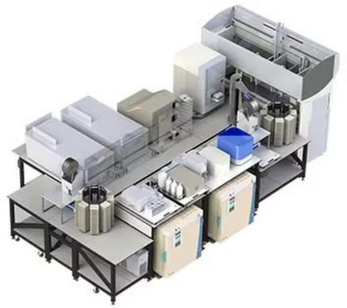
→ Screening
→ Sample prep

Configurable platform journey

Need to automate multiple experiments?

Key technologies & workflows:

→ Immunoassays
→ QA/QC testing

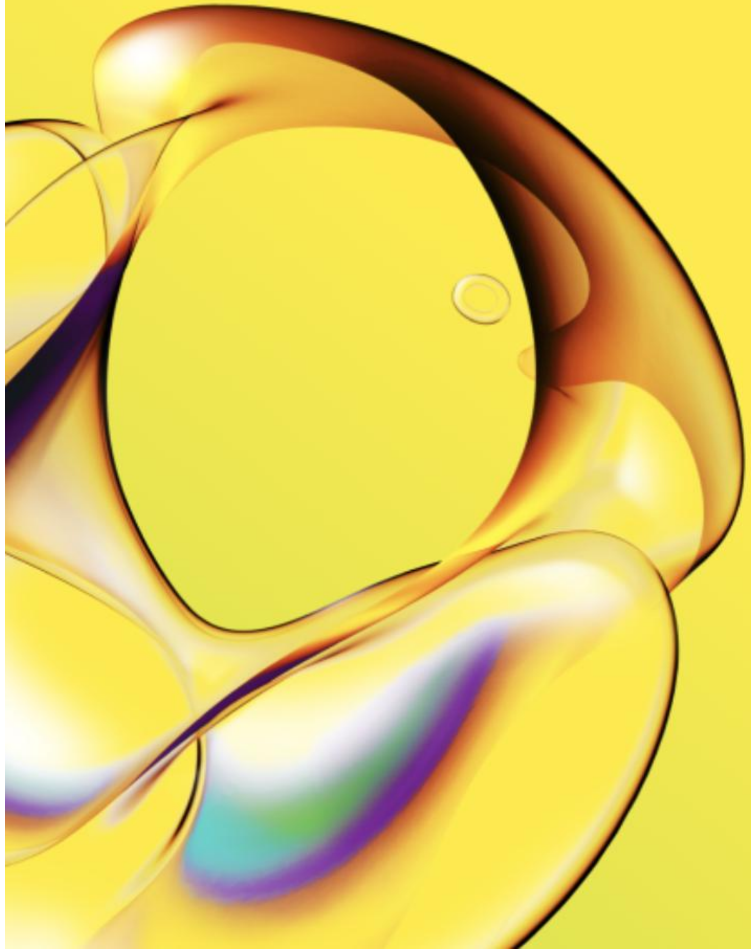


Large system journey

Need to automate a variety of complex interconnected processes?

Key technologies & workflows:

→ Protein expression
→ Cell line development



Automated instrumentation for superior success rates

- Unlock access to advanced technology that meets any laboratory need, from small size to high throughput
- Minimize time to market: automate steps, integrate data, streamline compliance, and more
- Reduce handling errors through robust readouts and advanced analytics

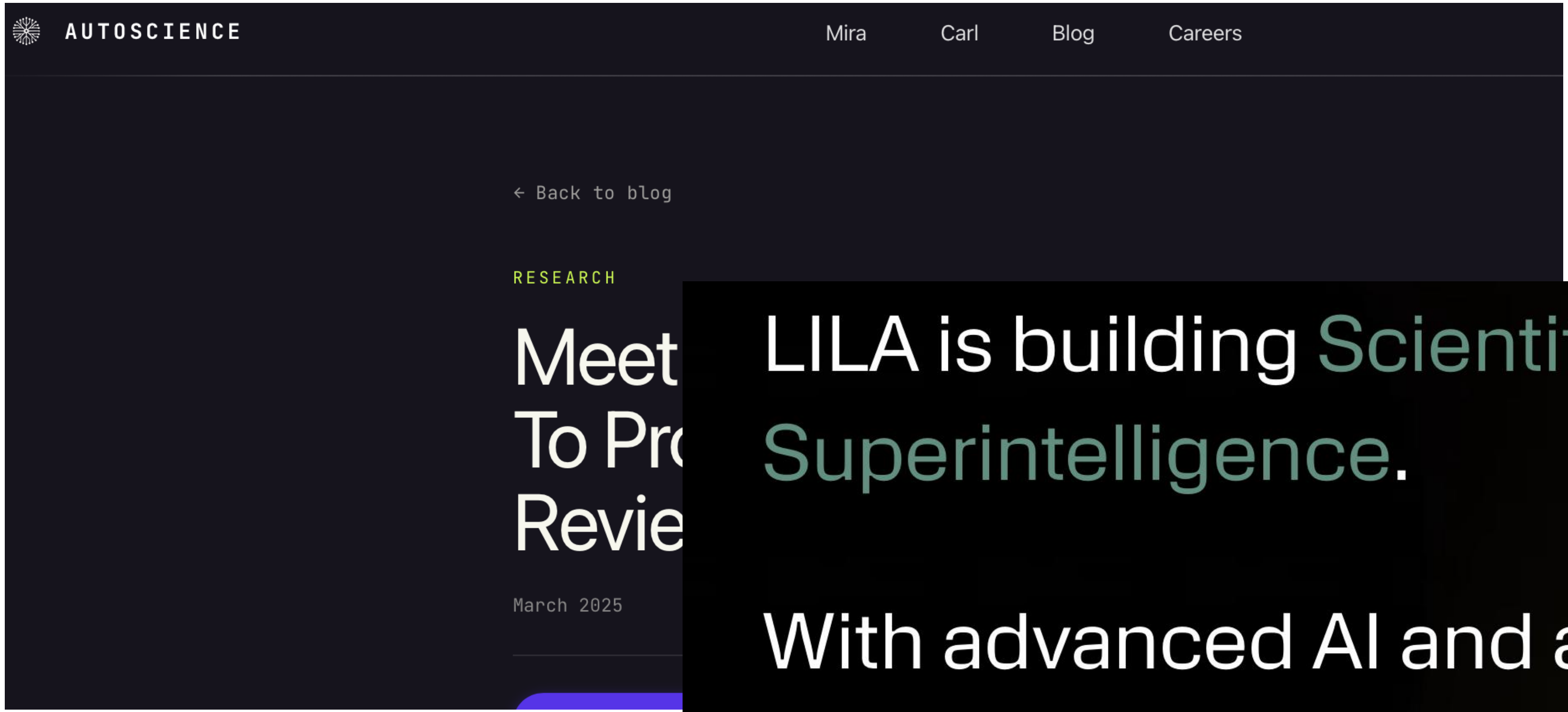
[formerly PerkinElmer]

[Emerald Cloud Laboratories]

Automation’s promise of a double-liberation

1. Free human researcher from “mere” manual labour (i.e., doing the experiment)
2. Free the experiment from human researcher (i.e., source of error)

- Presented as **win-win**: researchers can do “proper” science (the hard thinking and all that) **and science becomes not only faster but also more reproducible**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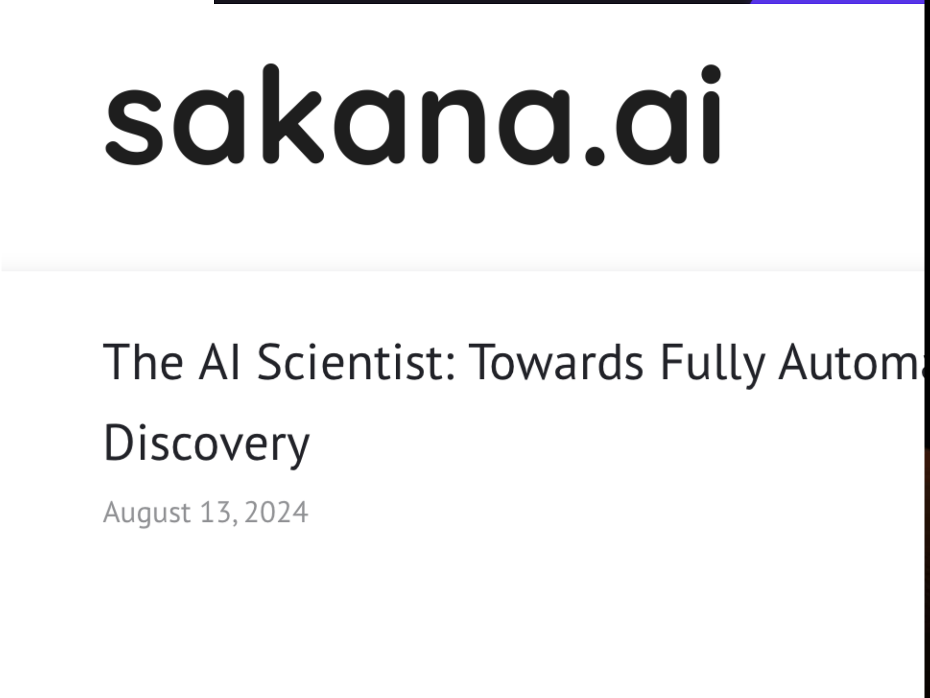
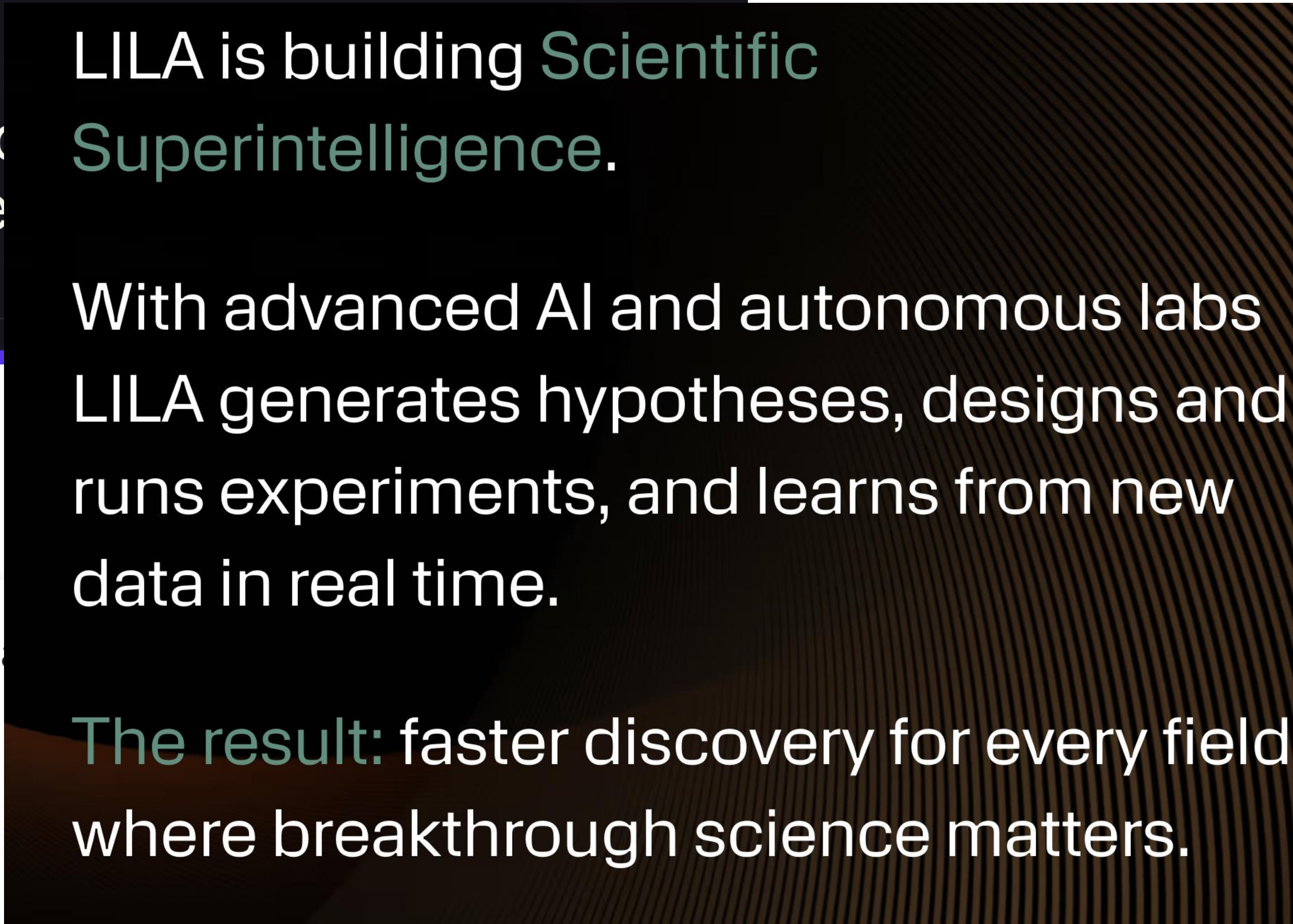

sakana.ai
The AI Scientist: Towards Fully Autom
Discovery
August 13, 2024


LILA is building Scientific Superintelligence.
With advanced AI and autonomous labs LILA generates hypotheses, designs and runs experiments, and learns from new data in real time.
The result: faster discovery for every field where breakthrough science matters.

Automation-*as-reform* and automation-*reformed*

1. Strong new push for more automation from academia and industry

2. Call for automation becomes a claim of *making-science-better* (rather than just economic proposition re speed or efficiency)

3. At the same time automation is *changing in type (ML-enabled)*
   → Further excitement, new opportunities

→*Automation is changing in type, speed, scale (fields) and depth (parts of field that are automated)*

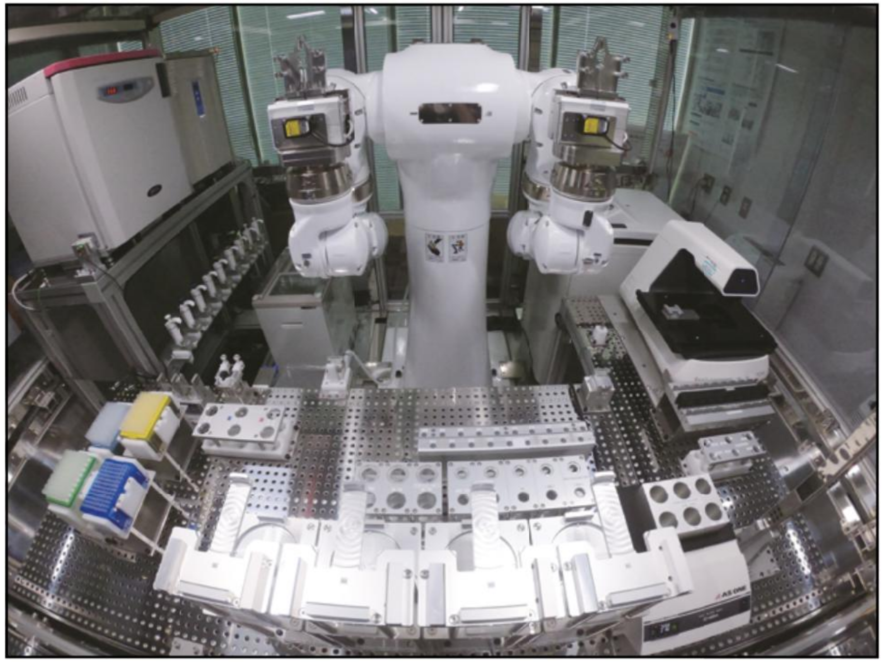
Three areas for critical reflection

How does the automation of research affect:

- The researcher?
- The research target?
- The research process/ecosystem?

What does “affect” mean?

- Epistemic impact (knowledge generation)
- Financial impact
- Legal issues (employment law, IP, etc.)
- Moral/ethical issues…



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Note: This is not just about AI. Automation comes as a larger package of changes

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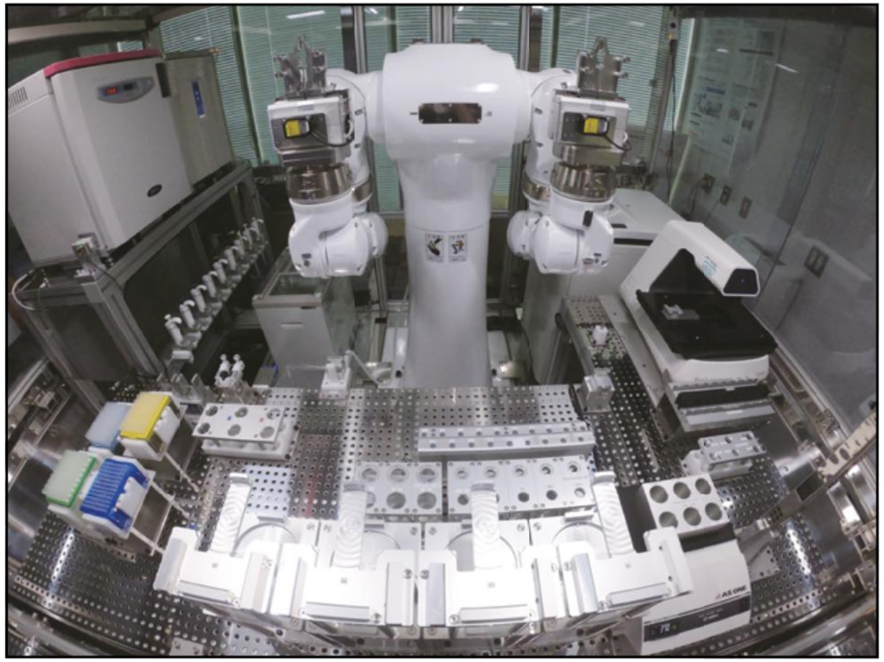

Three areas for critical reflection

How does the automation of research affect:

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Can a researcher be affected/harmed by automation in a morally relevant sense?

on)

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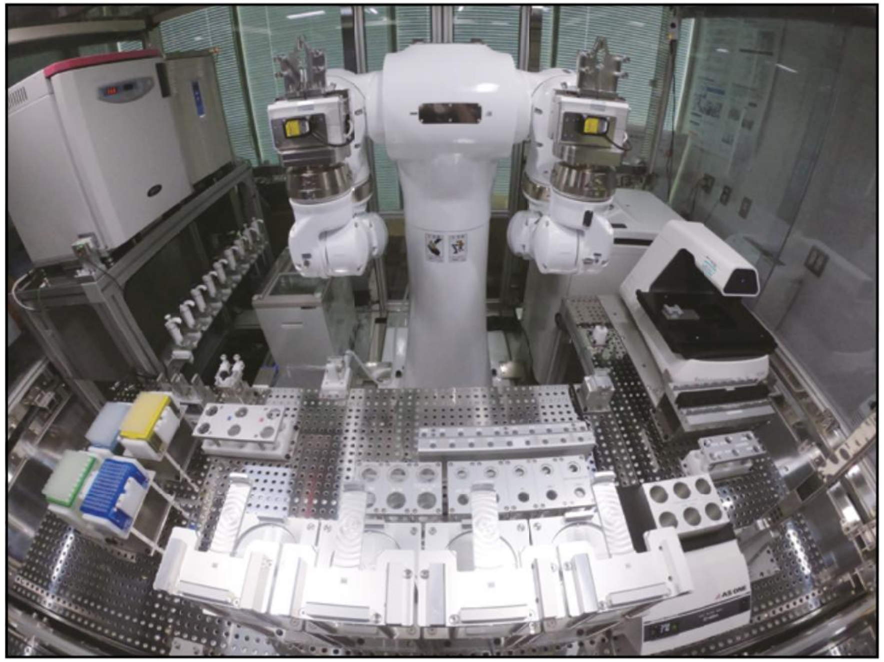

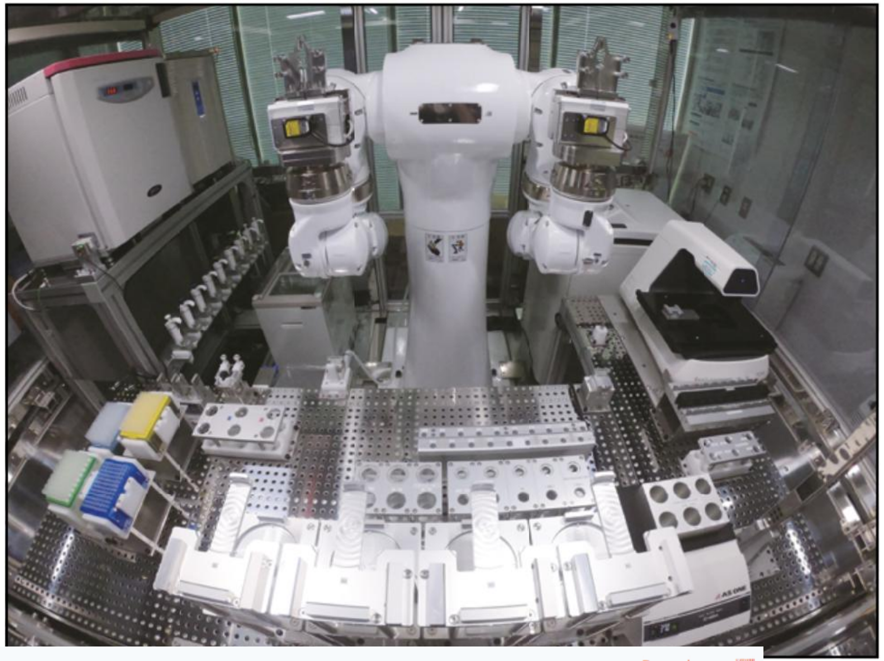

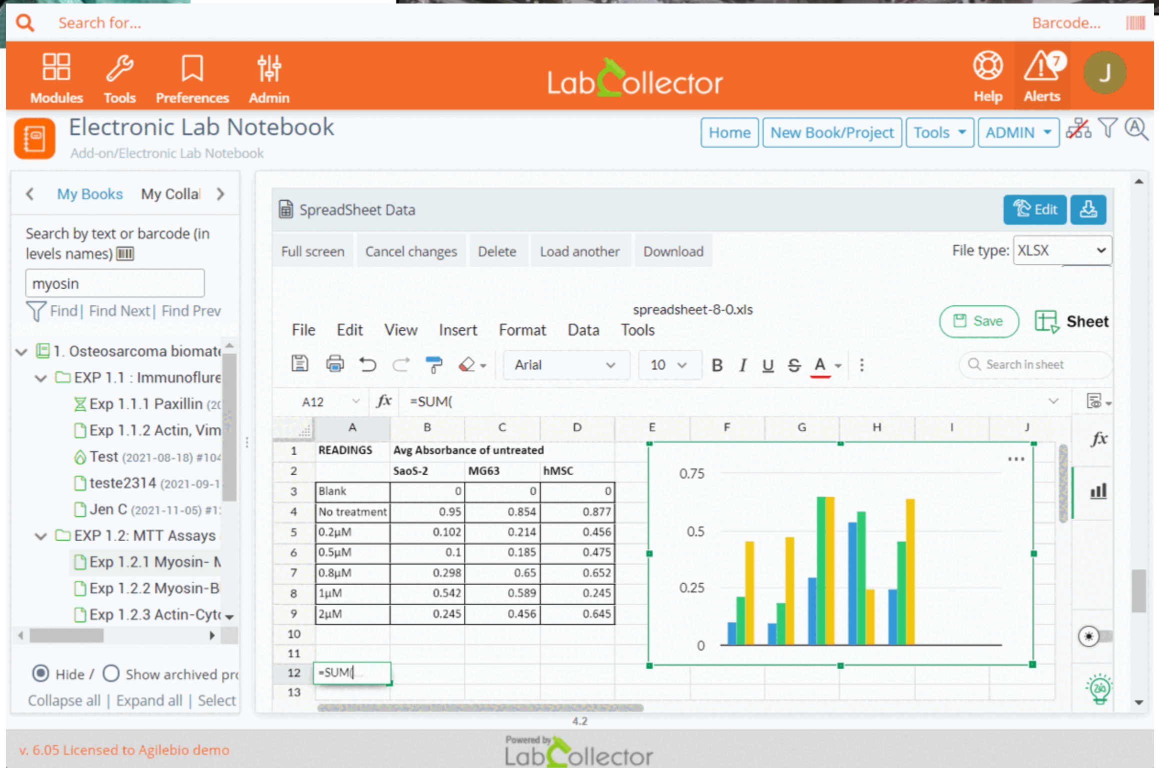
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Why automate note-taking?

SciNote

Product Solutions Reviews Plans Resources Company Log In GET A DEMO

Advantages of using an electronic laboratory notebook software

An ELN software will save you time

Quickly find any research data in your digital lab notebook, replicate processes and experiments with templates, and report faster with auto-generated project reports. SciNote ELN users save on average 9 hours per week.

Fast return on investment (ROI)

An ELN lab notebook software improves efficiency, collaboration, and data security, resulting in faster and more accurate data recording, simplifying workflows and making data sharing easier. It helps labs achieve their research objectives quickly and cost-effectively.

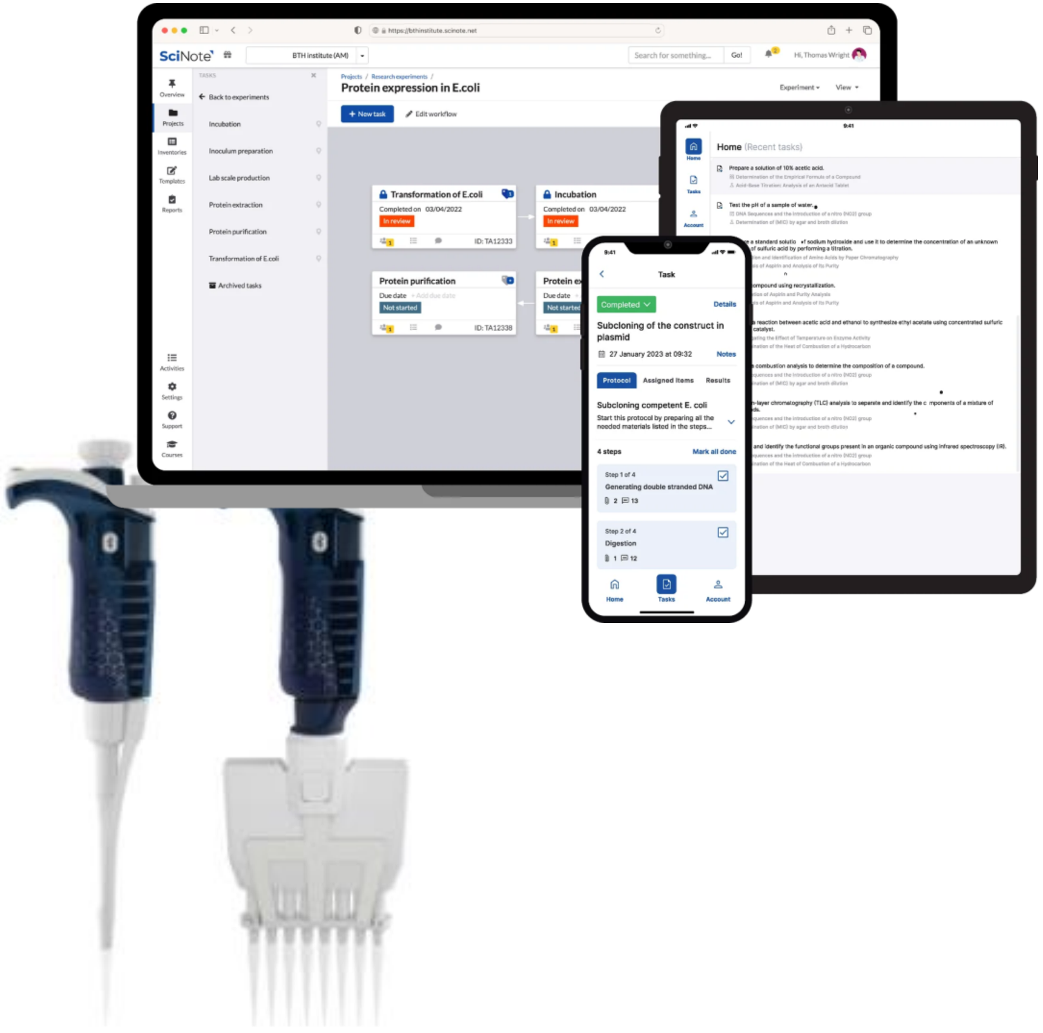
It's easy to learn and a pleasure to use

SciNote adopts a projects / experiments / tasks structure that emulates how research happens, and a flexible set-up you can customize to fit your needs. You can even see how connected tasks fit together through visualized workflows.

Improve your research data reliability and protection level

Auto-backup your scientific data and incorporate data integrity best practices into your day-to-day activities for GxP and 21 CFR Part 11 compliance. A digital laboratory notebook implements high-level security measures, so your data are well-protected.

ELNs and even whole laboratories via “Cloud”



CER: Cloud-enabled research

Benefits of automated note-taking and experiments

Make your records…

- …more searchable
- …more complete
- … easily shareable
- …better integrated with various (smart) platforms
- …

Make your research…

- …more efficient (run experiments in parallel, 24/7, etc.)
- …cheaper (no need to buy expensive tools – rent machines in ECL)
- …more reproducible (fewer pipetting errors, mis-takes, etc.)
- …traceable, accountable (patents, licencing)
- …

Clear benefits, but what are downsides of CER?

- Does it increase or decrease broader access to “cutting-edge” research tools? Emerald Cloud Lab claims the former…
- Surveillance at the workplace (every movement tracked)?
- Privileged access to research data (monopoly?)
- Epistemic-methodological challenge: experimental design?

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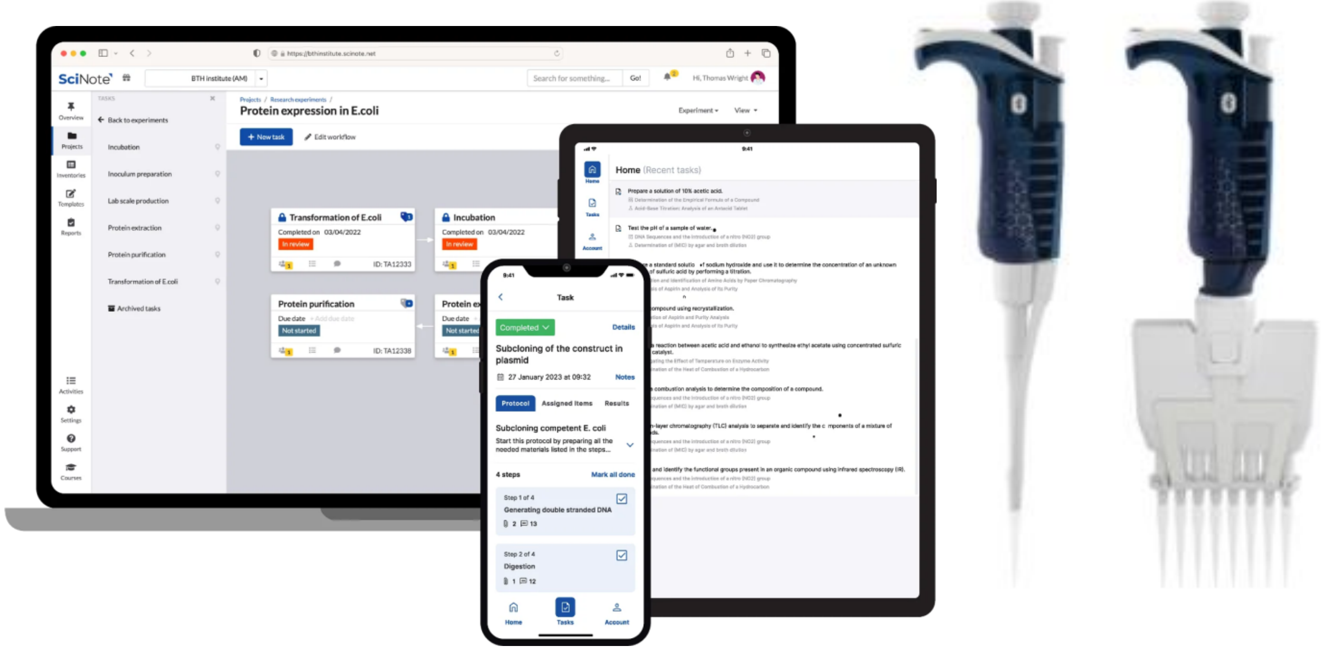
Why (semi-)automate data collection?

Cloud-based ELN (SciNote; collaborating with Gilson)



Make Your Lab as Smart as Your Science

Reduce Human Errors and Improve Traceability with TRACKMAN® Connected

How much data is enough/relevant?

Data collected by connected Gilson pipette/SciNote

- When and where did researcher X open their notebook and start work?
- What instrument did they use, connected to which protocol?
- How much time did they spend pipetting, at what temperature and relative humidity?
- What experimental design did they choose, and what reagents did they use to perform the experiments?
- What file types did they upload and how and when did they share them with other researchers?

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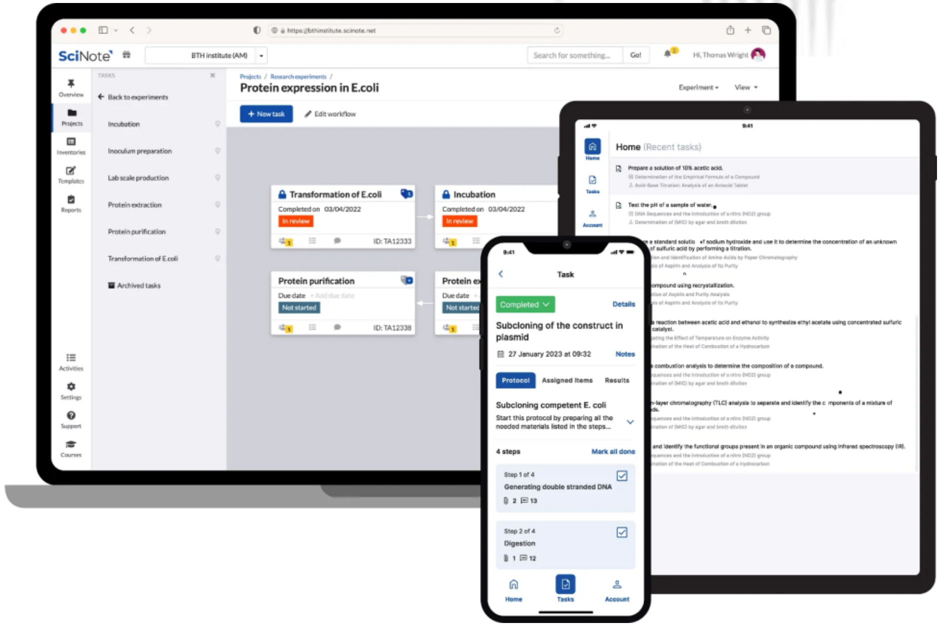


Laboratory privacy: Access to research content?

From Emerald Cloud Lab (ECL) “Legal” section:

- 2.2. Content Privacy. Unless otherwise specified by you or another ECL User

11. Access to Workspace

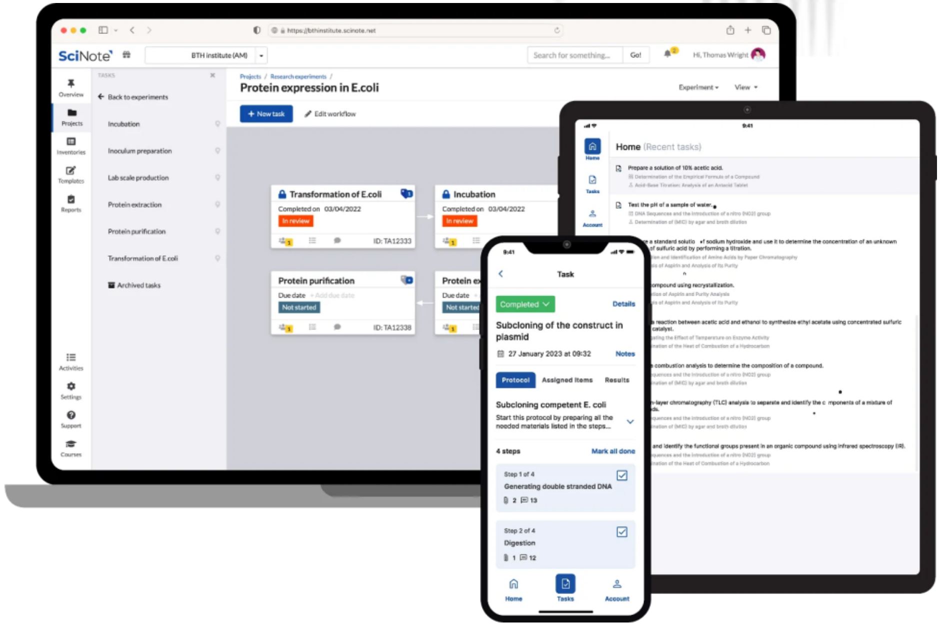
11.1. SciNote may view, analyze or otherwise process Customer’s Research Data and Activity Data for any of the following purposes: (a) provision of Services to the Customer; (b) maintenance and improvement of Services, including of SciNote ELN; (c) system administration; (d) support; (e) product and service improvements; (f) performance any other obligations under the Contract; (g) compliance with any applicable laws and regulations; or (h) enforcement of its rights under the Contract.

and any other instructions or materials submitted to or created in the ECL to ensure the safety, reliability, and performance of the ECL, to troubleshoot an Experiment to diagnose any potential errors, as well as to derive information to improve or augment any aspect of the ECL.

Cloud-enabled research – the Gilson pipette

Gilson/Scinote can gain more fine-grained picture of work in specific lab than PI themselves!

- Does third-party (instrument supplier, service provider) have the right to such detailed picture of worker's practice?
- Does potential for surveillance of employee by PI/institution affect experimental work itself?
    - Currently: each researcher has their own little research space – let their PI "in" at update meetings ("what have you done, where are we with this experiment…?" etc.). Social processes of lab remodelled → Relevant change?



The Panopticon lab?

Automated science and privacy/surveillance

- Automation of research comes with a range of changes – *not just about AI*
- One example: extensive data collection for **tracking of research activity through cloud-enabled research tools**/processes
- Pro: Benefits for replicability of research
- But: Risk of extensive surveillance of researchers
- Can researchers opt out? Should they be able to?
- Should there be some level of privacy in workplace?
- Also: epistemic-methodological issues raised by move to CER

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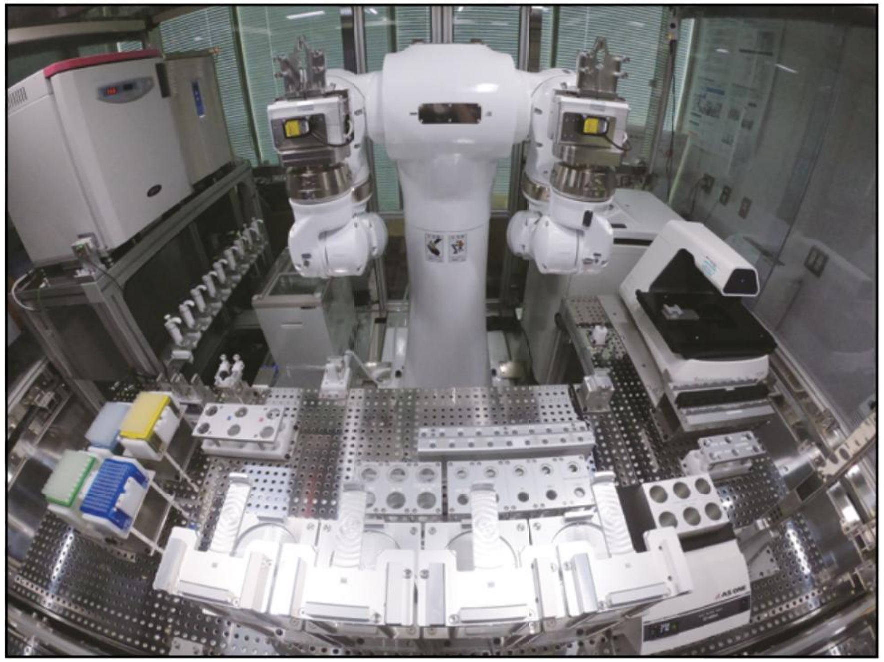

What does "affect" mean?

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Research Ethics – a quick word on ethical frameworks

- You are asked by your supervisor to design a new research plan for your PhD project which deals with weight loss drugs

- You want to make sure you choose an ethically acceptable course of action

- ***How do you even start to think about what might be morally right or wrong about your actions? What framework would you apply?***

Research Ethics – a quick word on ethical frameworks

Some of the big frameworks:

- Consequentialism (e.g., utilitrarianism)
- Deontology (focus on autonomy, includes rights-based approaches)
- Principlism (a mix of the above and more)
- Virtue ethics (focus on character of individual)
- Relational ethics

Research ethics

UKRI **principles**:

- research should aim to maximise benefit for individuals and society and minimise risk and harm [Consequentialism]
- the rights and dignity of individuals and groups should be respected
- wherever possible, participation should be voluntary and appropriately informed [Autonomy]
- research should be conducted with **integrity and transparency**
- lines of responsibility and **accountability** should be clearly defined
- independence of research should be maintained and where conflicts of interest cannot be avoided they should be made explicit.

Research integrity and accountability

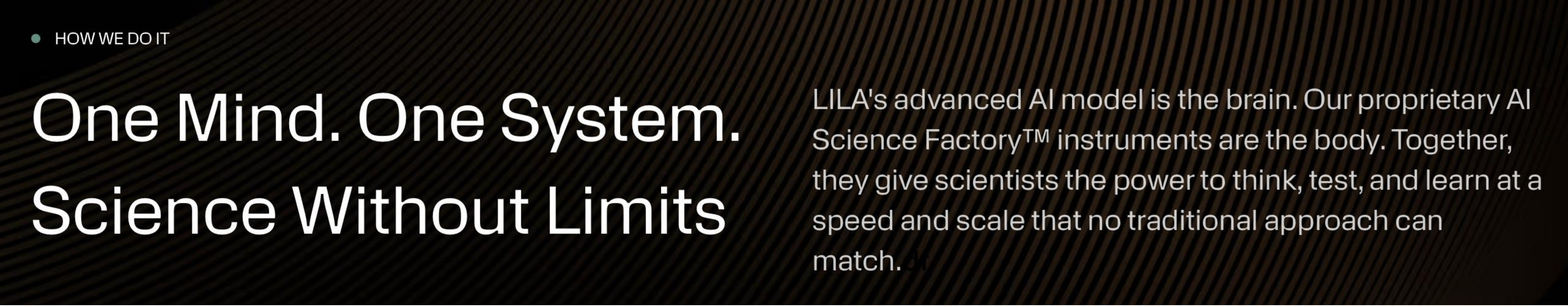
- Requires **authorship/agency** and **autonomy** (person conditions)

  We don't blame someone for lack of integrity and we don't hold them accountable if they were not (co-)author of a text or output

  We also withhold moral judgment if person acted under duress (forced labour)

  We DO hold people accountable even if they did not fully understand a process – they could have informed themselves or not engaged in process…

- **Transparency** and **explicability** are necessary to hold people accountable (process conditions)



Challenge: What does research integrity look like in the age of AI agents and automated science? Who do we hold accountable when things go wrong?

- Machines usually not seen as part of moral community, only “persons” are
- Developer of a tool might be partly responsible, but cannot fully control what others do with that tool
- Technical safety features might not always work, and will only work for anticipated threats/risk

Technology

All major AI models risk encouraging dangerous science experiments

“Google, DeepSeek, Meta, Mistral and Anthropic did not respond to a request for comment. “
https://www.newscientist.com/article/2511098-all-major-ai-models-risk-encouraging-dangerous-science-experiments/

https://www.nature.com/articles/s42256-025-01152-1

Group discussion

How can we ensure that automated research is ethical and has integrity?

- Should we oppose the rise of AI Scientists?
- Should we challenge the ”move fast break things” Silicon Valley mentality?
- Or should we just trust the process and stop worrying?

LILA is building Scientific Superintelligence.

With advanced AI and autonomous labs LILA generates hypotheses, designs and runs experiments, and learns from new data in real time.

The result: faster discovery for every field where breakthrough science matters.

Research integrity and accountability

Human-in-the-loop (HITL) refers to a system or process in which a human actively participates in the operation, supervision or decision-making of an automated system. In the context of AI, HITL means that humans are involved at some point in the AI workflow to ensure accuracy, safety, accountability or ethical decision-making.

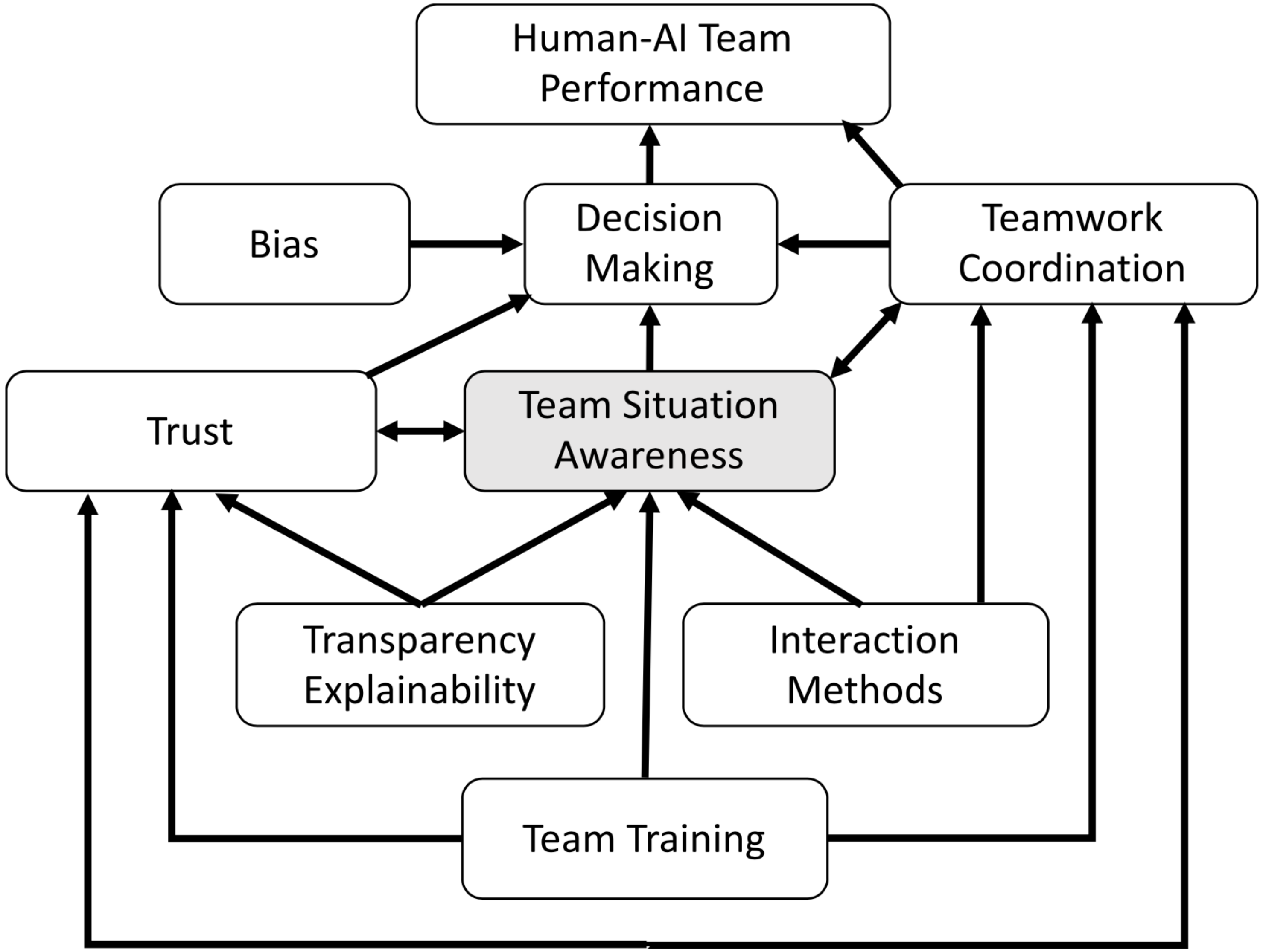
https://www.ibm.com/think/topics/human-in-the-loop

- One benefit: If things go wrong, we know who to hold accountable. **But is this fair to the human who is put in that loop?**

How should we think about the relation between human and machine?

Table 1. Summary of the widely cited, but mind-limiting 1-dimensional Sheridan-Verplank levels of automation/autonomy (Parasuraman et al., 2000).

Level	Description
High	10. The computer decides everything and acts autonomously, ignoring the human.
	9. The computer informs the human only if it, the computer, decides to.
	8. The computer informs the human only if asked, or
	7. The computer executes automatically, then necessarily informs the human, and
	6. The computer allows the human a restricted time to veto before automatic execution, or
	5. The computer executes that suggestion if the human approves, or
	4. The computer suggests one alternative, or
	3. The computer narrows the selection down to a few, or
	2. The computer offers a complete set of decision/action alternatives, or
Low	1. The computer offers no assistance; the human must take all decisions and actions.



Endsley 2023

Does the human-in-the-loop have sufficient situation awareness?

Three (related) reasons for doubt:

- Opacity of ML models (“Alien Intelligence”)
- Complexity of research pipelines, phenomena, and insights
- Novel experimental technology designed for AI

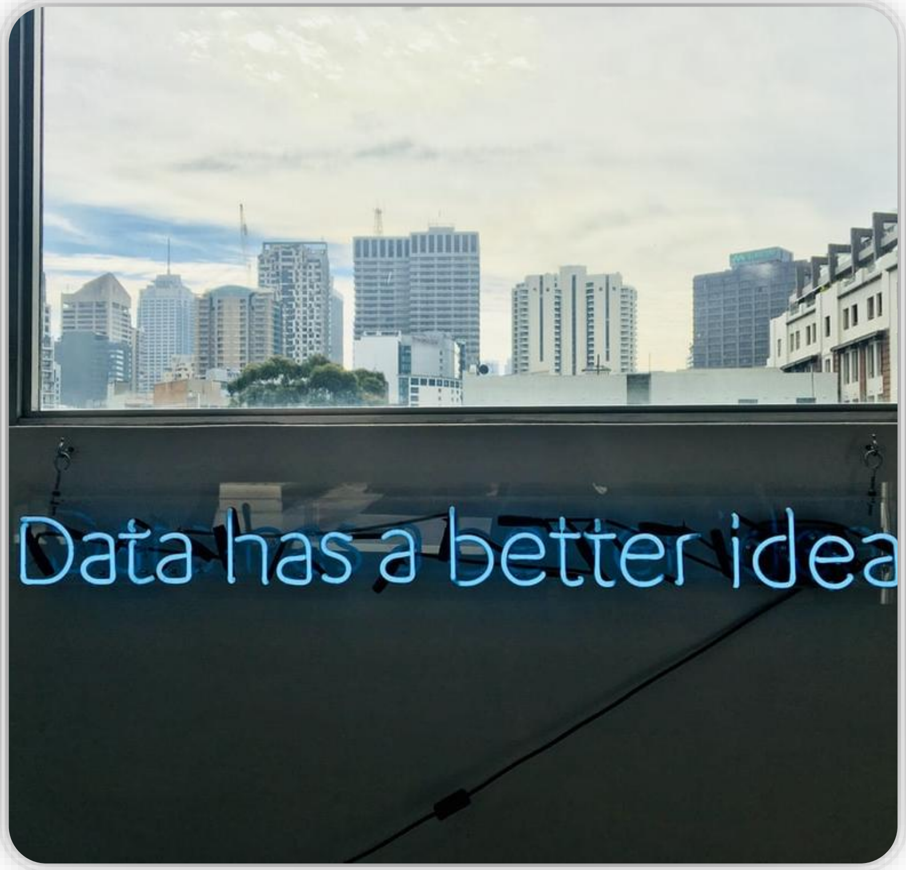
→ Lack of human understanding?

Weinberger (2017): The challenge of "alien" intelligence (AI)

1. Humans have limited cognitive capabilities
2. We shaped our tools/models accordingly
3. Tools work if we assume a simple world

But: World is not simple?

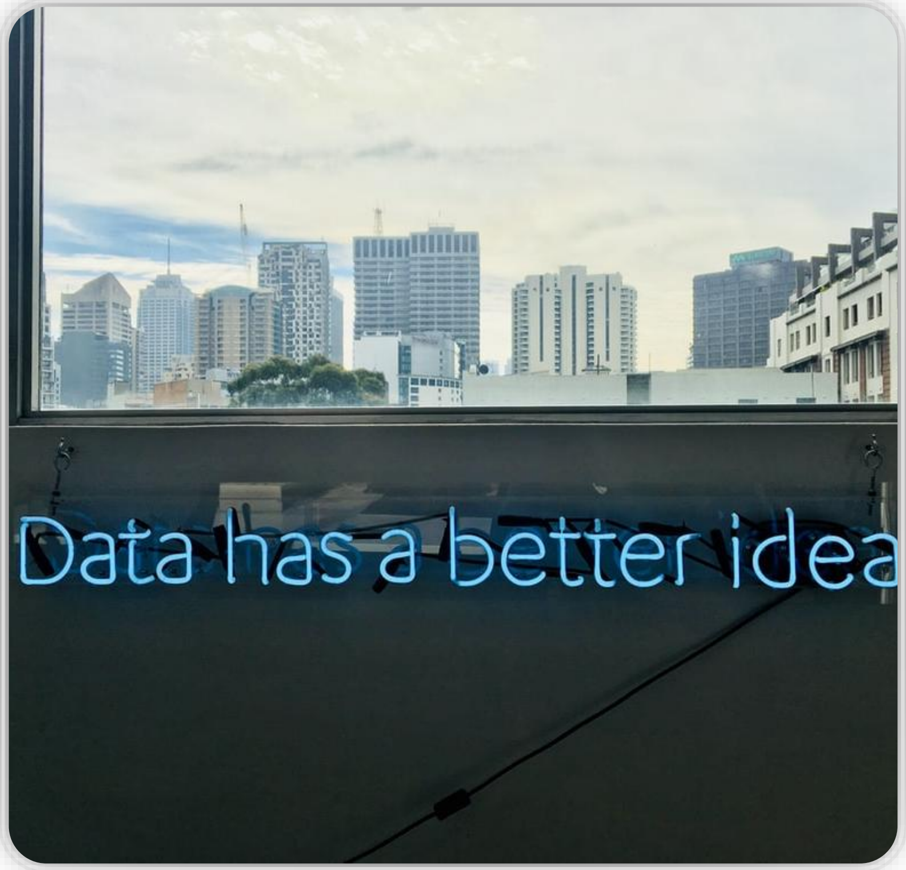
Maybe machines are better equipped to deal with the complexity of the world?



Weinberger (2017): “Alien” does not mean “wrong”

“Rather than reducing phenomena to fit a relatively simple model [which is what we used to do], we can now let our computers make models as big as they need to”

New epistemic-methodological situation: “never before have we relied on things that did not mirror human patterns of reasoning”



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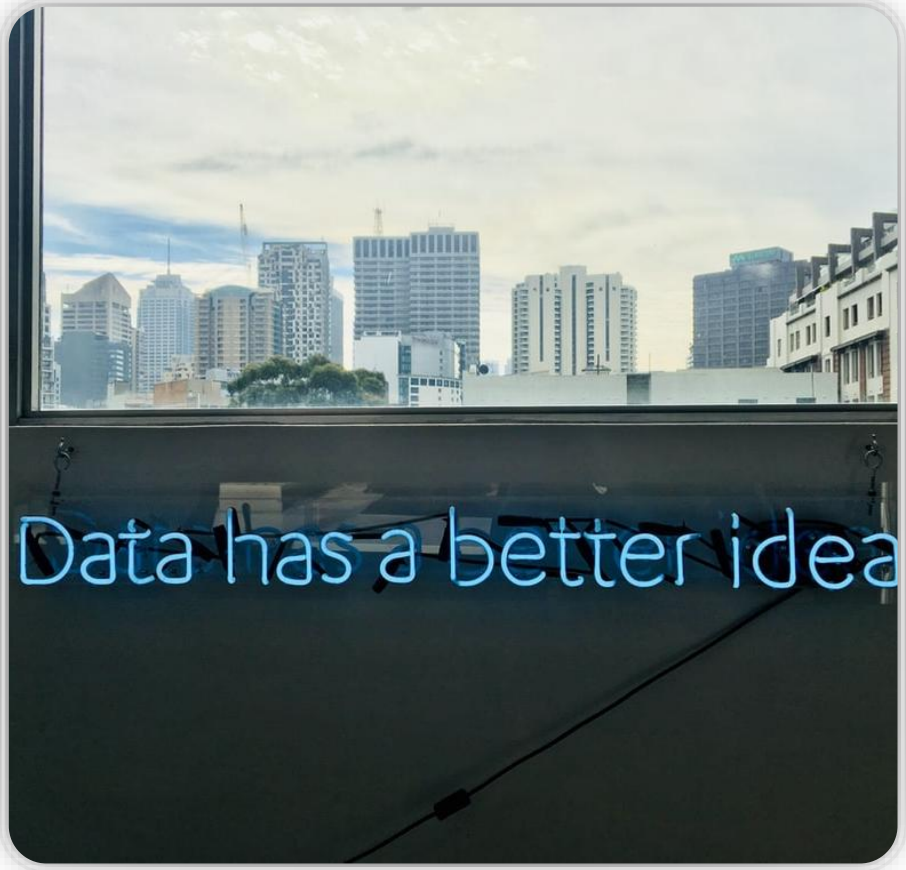
Downside: Infusion of "alien knowledge"

Machines derive conclusions from models we cannot understand.

"We thought [knowledge] was about simplifying the world. It looks like we were wrong. Knowing the world may require giving up on understanding it"



The challenge of AI-driven discovery

Nickles 2020:

- Mechanisation of discovery through machine learning: “emerging post-human style of reasoning”.
- Models produced by machines are opaque →New methods challenge human Intelligibility
- Implications? What role/value left for human modes of reasoning?

Nickles 2020: Nothing new?

- Science always underwent big transformations (theory and methodology) and human intelligibility would often be challenged as a result (e.g., quantum physics)
- “Alien minds” not something new: other animals and our own mind!
  - We don’t/can’t understand basis of scientific discovery, “creativity”, or ”skill” → Competence without comprehension
- Also: Science will not suddenly be “human-free”

  Will always reflect our cognitive limits, interests and purposes etc.

Nickles 2020:

- Problem of opacity of algorithms → Lack of critical scrutiny and hence accountability
  - Can we still detect errors and problems in these systems?
  - Human modes of reasoning will have decreasing value in fields where AI takes over. Human experts will struggle to identify or anticipate weaknesses in models in these fields
- Combined with problematic assumptions about Big Data as a reliable and unbiased evidence base for computational analysis
  - Biased data + AI has potential to create new inequalities that we might not be able to anticipate

Nickles 2020:

- Does “post-human style of reasoning” promise progress?

  Yes, in sense of extension of expert ***know-how***

- Does it cause problems?

  Yes. Risk of detrimental effects that cannot be anticipated due to lack of intelligibility (e.g., racialized facial recognition systems)

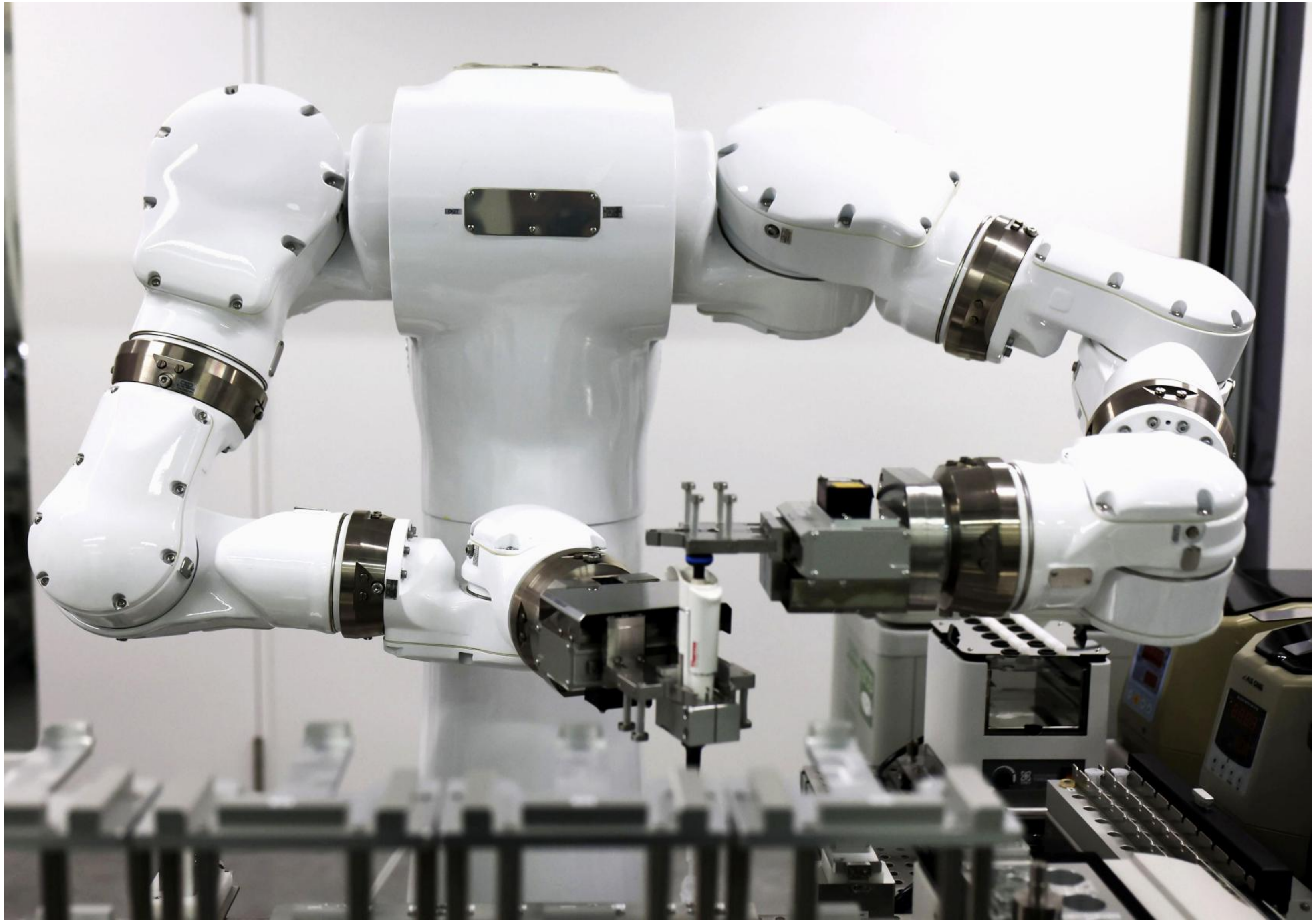
Weinberger (2017) and Nickles (2020):

- Machines give us “know-how” – predictions and correlations that help us do stuff without deeper understanding

- Nickles (2020): Scientific progress means ***expanding know-how of experts***

→ Deep problems for situation awareness of HITL?

Additional problem: tools designed for AI



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- **Transparency** and **explicability** are necessary to hold people accountable (process conditions)

Research integrity and accountability

What is the researcher's situational awareness in an automated workflow that is

- driven by an opaque AI model that
- uses tools not designed for human use and which
- produces models that are too complex for us to fully understand?

Does the HITL have enough agency is such a case to be held accountable?

How do we design HITL systems to make sure we are justified in holding people "in the loop" accountable?

The way forward: Reimagining what research looks like

- Automation of science is not just about which AI model to use and what capabilities this model (or multi-agent system) has
- Automation displaces human: it changes role/agency and responsibility of human. This affects situation awareness
- Large companies are driving change based on “faster discovery” narrative
- But how can we have a science in which research is done in an ethical manner? ***What spaces do we need to preserve or create for humans in this process? Bottom-up and local process?***